Weakly Supervised Semantic Segmentation in Aerial Imagery via Explicit Pixel-Level Constraints

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***Abstract*— In recent years, image-level weakly supervised semantic segmentation (WSSS) has developed rapidly in natural scenes due to the easy availability of classification tags. However, limited to complex backgrounds, multicategory scenes, and dense small targets in remote sensing (RS) images, relatively little research has been conducted in this field. To alleviate the impact of the above problems in RS scenes, a self-supervised Siamese network based on an explicit pixel-level constraints framework is proposed, which greatly improves the quality of class activation maps and positioning accuracy in multicategory RS scenes. Specifically, there are three novel devices in this article to promote performance to a new level: 1) a pixel-soft classification loss is proposed, which realizes explicit constraints on pixels during the image-level training; 2) a pixel global awareness module, which captures high-level semantic context and low-level pixel spatial information, is constructed to improve the consistency and accuracy of RS object segmentation; and 3) a dynamic multiscale fusion module with a gating mechanism is devised, which enhances feature representation and improves the positioning accuracy of RS objects, particularly on small and dense objects. Experiments on two RS challenge datasets demonstrate that these proposed modules achieve new state-of-the-art results by only using image-level labels, which improves mean Intersection over Union (mIoU) to 36.79% on iSAID and 45.43% on ISPRS in the WSSS task. To the best of our knowledge, this is the first work to perform image-level WSSS on multiclass RS scenes.**

***Index Terms*— Dynamic multiscale fusion (DMSF) mechanism, explicit pixel-level constraints, image-level weakly supervised semantic segmentation (WSSS), pixel global awareness (PGA), pixel soft classification (PSC), self-supervised.**

I. Introduction

AS A FUNDAMENTAL and challenging task in remote sensing (RS) image interpretation, semantic segmentation has become one of the most popular and important research fields. Though fully supervised learning approaches, such as ASPP [1], PSPNet [2], and DeeplabV3+ [3], have achieved significant success, their application scenarios remain limited because they rely on extensive manual pixel-level annotations. In comparison with fully supervised annotations, weakly supervised annotations are more efficient since fewer annotations are required, and fewer errors are likely to be made during manual labeling. Unlike fully supervised semantic segmentation (FSSS), weakly supervised semantic segmentation (WSSS) does not require pixel-level annotations during training, but it can get results that are comparable to FSSS performance, which is a significant advantage.

At present, most of the latest weakly supervised networks need class activation maps (CAMs) to generate pseudo-masks and then train segmentation models. However, the only available supervisory information is the category label in traditional classification models, which means that only the most discriminative parts of the CAMs can be activated during classification. Some efforts have been made to gain extra supervisory information from other existing single-class datasets [4], [5] or cheap web-crawled datasets [6], [7]. Some methods attempt to add more supervisory information to facilitate the activation degree of the target areas, such as SEAM [8], which utilizes equivariant constraints to guide online learning. Although researchers have made a number of breakthroughs in natural scenes (NSs) that have greatly improved the performance of image-level WSSS [8], [9], [10], [11], [12], there is less research on this task in the RS scene. The traditional methods [13], [14] are limited in RS by the multicategory, large-scale scenes, as well as dense small targets. In such a scenario, there are some unique challenges and problems to be solved.

1) Ultrarange Activation: In complex RS scenarios, multicategory scenes account for the vast majority. As shown in Table I, the single-category scenes of the PASCAL VOC 2012 account for 64%, whereas the single-category scenes of RS datasets account for less than 30%; especially for the

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TABLE I

STATISTICAL TABLE OF THE DISTRIBUTION OF IMAGE SCENE CATEGORIES FOR DIFFERENT DATASETS, WHICH RECORDS THE PROPORTION OF IMAGES WITH DIFFERENT CATEGORY SCENES TO ALL IMAGES IN THE ENTIRE DATASET, AND N IS THE NUMBER OF CATEGORIES IN A PICTURE. THE PERCENTAGES IN EACH COLUMN SHOW THE PROPORTION OF IMAGES IN THE ENTIRE DATASET AT DIFFERENT NUMBERS OF IMAGE SCENE CATEGORIES

Class number	PASCAL VOC 2012	ISPRS Vaihingen	iSAID
n = 1	63.98%	0.00%	27.37%
n = 2	25.88%	0.90%	42.69%
n = 3	8.35%	2.02%	20.92%
n = 4	1.59%	25.56%	7.20%
n = 5	0.21%	64.24%	1.31%
n = 6	0.00%	7.20%	0.40%
n = 7	0.00%	0.00%	0.11%

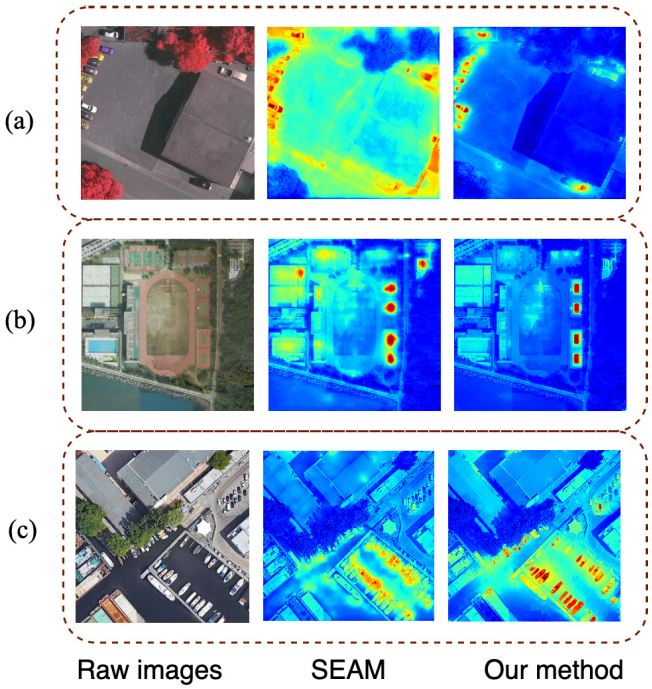


Fig. 1. Some specific matters in the RS scenes. (a) Ultrarange activation: the CAMs in complex multicategory RS scenes easily fall into local minimums, resulting in ultrarange activation in the background region. (b) Mismatched target edges: in CAMs, the semantic boundary of the RS target does not match its geometric boundary, which means that it cannot satisfy the segmentation mask requirements. (c) Rough target location: a large number of small targets and densely distributed targets in RS scenes cannot be accurately located.

ISPRS Vaihingen, the single-category scenes account for 0%. The existing NS classification model only uses image-level constraints to deal with pixel-level problems in multicategory scenes, which is not enough. During training, insufficient constraints produce a trivial solution. As shown in Fig. 1(a), the generated CAMs easily fall into local minimums and cause ultrarange activation in the background area. Ultrarange activation refers to the phenomenon that a large part of the area in a picture is activated. To overcome the problem of insufficient constraints, a novel pixel-level loss and an encoding mechanism are proposed to provide explicit constraints for each pixel, which successfully transfers the classification supervision from the image level to the pixel level to greatly avoid ultrarange activation.

2) Mismatched Target Edges: Unlike in NS images, the distribution of the foreground and background in the RS images is extremely imbalanced. As shown in Fig. 1(b), when directly applying existing methods, it tends to introduce redundant background information and incorrect interclass information, resulting in an obvious mismatch between semantic edges and geometric edges. To solve this mismatch issue, we utilize the low-level visual information of the features to adjust the semantic boundaries and the high-level semantic information to enhance the consistency of the activation regions within the target. Our method aligns the semantic boundaries of the targets with the geometric boundaries to obtain more precise and clear semantic boundaries.

3) Rough Target Location: There are many small-scale dense objects in RS scenes, which are rarely present in the traditional NS scene. Although lots of feature fusion methods [12], [13], [14] have been proposed to enhance the feature representation ability of models, these methods introduce a lot of redundant information, as shown in Fig. 1(c), resulting in a large number of small-scale dense objects to be submerged in the background and thus indistinguishable. In response to the aforementioned issues, a multiscale feature adaptive fusion module has been devised to effectively reduce the redundancy in low-level features and background noise, thereby improving the accuracy of RS object localization, particularly for tiny and dense objects.

In summary, the main contributions of this work are given as follows.

1) We propose a self-supervised Siamese network based on the explicit pixel-level constraints (S^2EPC) framework as the first benchmark for image-level WSSS tasks on multiclass RS scenes.
2) A pixel-soft classification (PSC) loss and pixel adaptive encoding (PAE) mechanism are proposed to generate explicit pixel-level constraints and enhance the fine graininess of CAMs in complex RS scenes.
3) A pixel global awareness (PGA) module is devised to align the semantic and geometric boundaries of RS targets, which improves the consistency of pixels within the class, along with the sharpness of the target boundaries.
4) To ensure accurate positioning of RS targets, a dynamic multiscale fusion (DMSF) module with a gating mechanism is employed to improve the feature representation of small-scale dense objects.

Compared with other related methods, the proposed method generates pixel-level constraints and makes CAMs in complex RS scenes much more fine-grained. Experiments on the iSAID [15] and ISPRS Vaihingen [16] datasets have demonstrated the effectiveness of our method in image-level WSSS tasks. Furthermore, we also conduct experiments on the PASCAL VOC 2012 [17] dataset, which verifies that the proposed method can also boost model performance in NS. Subsequently, we briefly introduce related work on image-level WSSS in Section II. Section III illustrates the details of the S^2EPC framework and specific algorithms, which include

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the PAE module, the PGA module, and the DMSF module. The experimental analysis and results on the iSAID, ISPRS Vaihingen, and PASCAL VOC 2012 datasets are presented in Section IV. Finally, the conclusion is given in Section V.

II. Related Works

A. Semantic Segmentation in Remote Sensing Scenes

In recent years, semantic segmentation methods based on deep learning had been increasingly applied to the processing of high-resolution RS images. Numerous fully supervised methods based on deep learning had achieved significant success in optical RS images due to the powerful feature representation capabilities of deep learning. Schuegraf et al. [18] used two parallel U-Net networks to extract elevation features and spectral features from RS images and then combined these two features to extract buildings. Niu et al. [19] proposed HMANet, an attention-based hybrid multiple attention network that adaptively captured global relevance from spatial, channel, and category perspectives.

The above approaches relied on pixel-level annotations, which require considerable time and effort. To reduce the burden of manual labeling of pixel-level labels, semisupervised, weakly supervised, and unsupervised methods gradually emerged in the field of RS segmentation. Existing mainstream semisupervised methods [20], [21], [22] used generative adversarial networks (GANs) to get close to fully supervised segmentation with few pixel-level labels. Sun et al. [23] proposed the boundary-aware semisupervised semantic segmentation network (BAS^{4}Net), a semisupervised semantic segmentation framework based on GANs, to accomplish the segmentation of very high-resolution RS images by inferring pseudolabels from unlabeled images. He et al. [24] designed a semisupervised learning method based on consistency regularization with a hybrid perturbation paradigm to improve the model performance in the road, building, and land cover datasets.

However, the above fully supervised and semisupervised methods still rely on pixel-level labels in essence. Some researchers had started to work on unsupervised segmentation methods in RS scenes. Grozavu et al. [25] proposed an automatic image segmentation method based on topology learning to segment images into different regions by different textures, intensities, and colors. Saha et al. [26] sampled smaller patches from a single scene and used deep clustering, spatial consistency, and contrastive learning to automatically segregate the major classes present in the scene and produce the segmentation map. Scheibenreif et al. [27] based on transformer architecture first performed self-supervised pretraining on large-scale unlabeled RS data and then fine-tuned the model on small labeled datasets. Among the several unsupervised methods above, the method that Saha et al. [26] proposed used the Siamese-like network structure without shared weights to reduce the gap between feature representations between multiple views of the same image. In contrast, our network shares weights from the backbone to all subsequent modules to reduce the gap between image-level annotations and pixel-level annotations. Some other methods chose other forms of labels with less supervision than pixel-level labels to achieve the segmentation task, which were all categorized as WSSS methods. In this article, our method also belongs to the WSSS methods, which is described in detail in Section II-B.

B. Weakly Supervised Semantic Segmentation

To relieve the burden of pixel-level labeling, numerous weakly supervised methods were proposed. According to different labeling types, these methods could be divided into the following three categories: scribbles [28], [29], [30], bounding boxes [31], [32], [33], [34], and image-level annotations [35], [36], [37], [38]. Specifically, scribbles used a limited number of pixels (dot or line) to provide information about the location and category of the target. Similarly, bounding-box annotations utilized geometric boxes to capture the corresponding target position and category information. Unlike the above types, image-level annotation was more challenging since it only provides category information, not location, and shape information.

As the field of WSSS develops, numerous advanced studies had utilized image-level labels to execute the segmentation task in NS, whereas there were few studies that focused on this task in RS. WSF-NET [13] created a feature fusion network to represent the target features of RS images, and a balanced binary training strategy was designed to eliminate the class imbalance problem. SPMF-Net [14] utilized the super-pixel pooling and multiscale feature fusion method to make detailed segmentation predictions of buildings. Both of them were binary segmentation frameworks, which only dealt with no more than two simple categories. Both of them only dealt with no more than two simple categories. In this article, we attempted to extend these tasks to multicategory complex RS scenes to fill the gap in the field of image-level WSSS.

C. Class Activation Maps

According to the best of our knowledge, most of the mainstream image-level WSSS algorithms utilized pseudomasks generated from CAMs to train the segmentation model, with class labels being the only available supervisory information. CAMs originally served as a visualization method to illustrate the working mechanism and underlying judgments of classification networks, which were created by weighting each channel. The area was darker; the higher the network response, the more significant the contribution. The CAMs generated by existing methods [39] usually covered only the most discriminative part of the target, and some false activations appeared in the background.

To address the above problems, much work focused on suppressing false activations in the background, which can obtain more complete and accurate CAMs. The existing methods for generating CAMs can be roughly categorized into two groups. The first group usually relied on complex networks and advanced pipelines to add additional information, which included saliency detection models [40] or video data [41]. However, under the situation of multicategory and complex backgrounds, the mask produced by saliency detection is not effective, and the introduction of extra datasets increased the complexity of model processing. The second group

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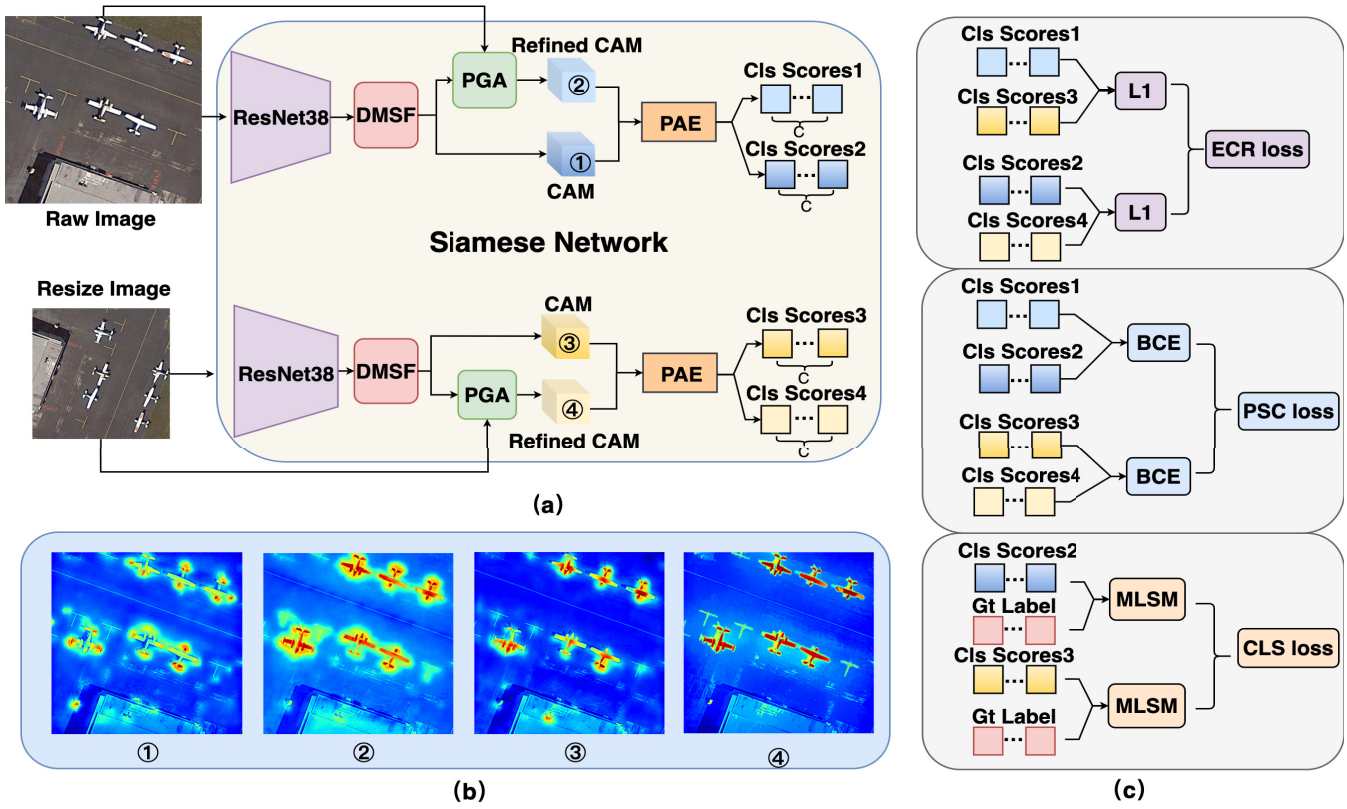


Fig. 2. (a) S^{2}EPC framework diagram, where the light yellow area represents the part of network weight sharing. S^{2}EPC contains three modules: the DMSF module, the PGA module, and the PAE module. (b) CAMs' visualization of two branches before and after refinement. (c) Schematic of the loss functions' calculation process. The CLS loss still adopts the MLSM loss as previous methods, and the ECR loss uses L1 loss to compare CAMs on two branches. The PSC loss is based on the binary cross-entropy (BCE) loss, which is used to calculate the pixel-level loss between the original CAMs and the PGA-optimized CAMs.

aimed to refine CAMs by guiding the classification network to capture more information during training. For example, Wei et al. [38] intentionally erased the target area by using an antagonistic erasure strategy and forced the network to find more discriminative information. For extending the attention of the network, Wei et al. [42] employed dilated convolution layers with different expansion rates. DRSG [43] applied seed region growth to improve initial location maps during training. However, neither of the aforementioned approaches could provide explicit pixel-level constraints during the training process.

In contrast to the aforementioned methods, we mined the supervision information in a self-supervised way to provide explicit constraints.

III. Methodology

Further information about the S^{2}EPC network is presented in this section. Section III-A shows the motivation of the problem setup, and Section III-B imports some details of the S^{2}EPC framework. Sections III-C–III-E introduce three essential modules: the PAE module, the PGA module, and the DMSF module. Section III-F mainly focuses on the loss function used in the S^{2}EPC network.

A. Problem Setup

As shown in Fig. 2, the S^{2}EPC network with the double-branch structure needs a pair of images $[I_0, I_1]$ as input, where $I_0 \in \mathbb{R}^{3\times H\times W}$ is the original image and $I_1 \in \mathbb{R}^{3\times H\times W}$ is the scaled image. In this part, semantic segmentation tasks are divided into two forms based on the different annotation types: the fully supervised task and the weakly supervised task.

For the FSSS task, there is

$$\mathrm{Seg}(I, Y_{\mathrm{gt}}) = Y_{\mathrm{seg}} \tag{1}$$

where the semantic segmentation model is denoted as $\mathrm{Seg}(\cdot)$ with the image I and the ground truth Y_{gt} as inputs, generating the predicted segmentation mask Y_{seg}. Similarly, the WSSS model can be expressed as follows:

$$\mathrm{Seg}(I, Y_{\mathrm{pseudo}}) = Y_{\mathrm{seg}} \tag{2}$$

where Y_{pseudo} is the pseudolabel for image segmentation. The goal of the WSSS task is to get pseudolabels Y_{pseudo} from the classifier $\mathrm{Cls}(\cdot)$. An ideal WSSS model should satisfy

$$\mathrm{Seg}(I, Y_{\mathrm{gt}}) = \mathrm{Seg}(I, Y_{\mathrm{pseudo}}) \tag{3}$$

that is, $Y_{\mathrm{pseudo}} \approx Y_{\mathrm{gt}}$. The majority of existing WSSS approaches rely on CAMs to generate the initial seeds for localization, from which CAMs are refined as pixel-level pseudolabels. After that, the pseudolabels are used as ground truth annotations to train the segmentation network. It is clear that the accuracy of the pseudolabels is crucial to the success of WSSS, and CAM is the primary factor influencing the performance of the pseudolabels.

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In the WSSS task, CAMs are also used to derive pseudolabels from mainstream methods. Prior to obtaining the multilabel CAMs, the foreground and background features are cascaded together so that the background pixels are taken into account. Equation (4) represents the process of calculating background features

$$\mathrm{Bg}(\boldsymbol{M}_{:,h,w}(x,y)) = 1 - \max_{1\le i\le c-1} \mathrm{M}_{i,h,w}(x,y). \tag{4}$$

Then, multilabel CAMs $M_{:,h,w}(x,y)$ are projected into single-label pseudolabels by using the argmax operation on the channel dimension, which can be expressed as follows:

$$\boldsymbol{Y}_{\mathrm{pseudo}} = \mathrm{argmax}(\mathrm{Cat}(\mathrm{Bg}(\boldsymbol{M}_{:,h,w}), \boldsymbol{M}_{:,h,w})) \tag{5}$$

where $\mathrm{Cat}(\cdot)$ represents the channelwise concatenation of the feature vectors. The CAMs are acquired based on the contributions of features in each category. Equation (6) describes the generation process of CAMs as follows:

$$M_{:,h,w}(x,y) = {W_c}^T \odot F_{:,h,w}(x,y) \tag{6}$$

where $F_{:,h,w}(x,y)$ denotes the feature of pixel at (x,y), W_c is the weight of the classifier, and $\odot$ denotes the elementwise multiplication.

In this article, our S^{2}EPC uses the SEAM [8] model as the baseline. To further improve the quality of CAMs, pixel-level explicit constraints are added to our network to iteratively optimize the CAMs and solve the image-level WSSS problem for complex RS scenes. Excellent CAMs not only need to be able to accurately locate targets but also satisfy semantic consistency and intraclass differences in pixels.

B. Siamese Classification Network

The S^{2}EPC network architecture is shown in Fig. 2, which consists of two inputs: the original image and the scaled image. The use of the double-branch Siamese network instead of the traditional single-branch network means that the network needs to generate equivalent activation areas in different sizes of the same image. This can be viewed as an approach of data augmentation, which can provide mutual supervision for the two branches without increasing the training samples. Furthermore, the pixel soft classification (PSC) loss is employed in the S^{2}EPC framework to enhance the fine graininess of CAMs in complex RS scenes and realize the self-supervised training on each branch. Compared to traditional classification networks, our two-branch S^{2}EPC network forms a double supervision mechanism to enhance the consistency and precision of network predictions.

1) Network Configuration: The design of the S^{2}EPC network [see Fig. 2(a)] is inspired by the SEAM [8] network. SEAM is based on the Siamese network architecture, which successfully applies the equivariant consistency constraint to the CAMs from different transformed images to provide self-supervised information for classification network learning. The Siamese network architecture is used to measure the degree of similarity between two inputs with shared weights, and the ResNet38 [10] network serves as the backbone. The ResNet38 network is a shallow residual network connected to the FCN, with few parameters and rapid training and inference speeds. In our network configuration, the last three convolution layers of the ResNet38 network are atrous convolution layers with dilation rates of 1, 2, and 4, respectively. The advantage is that the resolution of the feature maps does not decrease simultaneously as the receiver field increases, which enhances the performance of our networks relying on the feature maps. It is worth mentioning that the backbone network can also be changed into other structures according to the usage scenarios. For example, the ResNet50 or ResNet101 network structure could also be utilized to achieve a higher feature extraction effect. The performance of our S^{2}EPC network improves as the number of backbone layers increases, but the corresponding model inference time increases as well. The part of comparative experiments on SOTA methods shows the effects of different backbone networks.

2) Network Architecture During Training Phase: During the training phase, the S^{2}EPC network receives two input streams: the raw input image I_0 and the image I_1 randomly scaled in the interval [0.5, 1). These two streams are combined into a pair of images $[I_0, I_1]$ as the input of the S^{2}EPC. First, ResNet38 extracts the features of the image pair $[I_0, I_1]$. After that, the DMSF module is utilized to adaptively and dynamically fuse the high- and low-level features $[X_h, X_l]$. Following that, the fused features X_{hl} are sent to a classifier for classification. In order to obtain CAMs, feature maps are weighted according to their contribution to the output of the classifier. To further enhance the fine granularity of CAMs, the original CAMs and fused features X_{hl} are refined by PGA to obtain the refined CAMs. In order to maintain the activation strength of the original CAMs, the PGA module is placed on another branch parallel to the original one. After the classifier has been trained, our model can eventually generate refined CAMs and pseudolabels for all samples.

3) Network Architecture During Testing Phase: Considering that the S^{2}EPC network is a Siamese network with shared weights, all that is required during the testing phase is to restore the weights of one branch. Given an invisible test image, the multiscale test pipeline is utilized to predict the CAMs at various image scales, thereby maximizing inference performance. In line with common practice, we use dense-CRF [44] as a postprocessing step to improve CAMs and the argmax operation to project the refined CAMs to pseudolabels. In terms of semantic segmentation, we follow [8] and [10] to employ the DeepLabV1 [45] to train pseudolabels and predict the final segmentation results.

C. Pixel Adaptive Encoding

Segmentation is considered a pixel-level classification task, but existing classification models only use image-level constraints to train the network in NS and cannot guide the pixel-level classification tasks. Due to the insufficient model constraint capability, the existing methods are only applicable to simple NS scenes, and there are limitations for RS datasets dominated by multicategory scenes. During the training process, insufficient constraints make the CAMs frequently fall into local minimums, as shown in Fig. 1(a), resulting in ultrarange activation in the background. To overcome the

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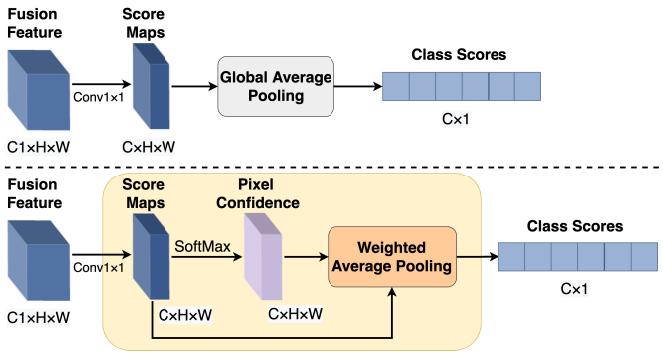


Fig. 3. Structure of the PAE module. The GAP layer is replaced with the weighted average pooling, and the weights are determined by the pixel confidence maps generated by the SoftMax function.

above problems, we design a novel pixel-level loss and pixel encoding mechanism, which are proposed to provide explicit constraints for pixels.

This section focuses on the pixel encoding mechanism. As illustrated in Fig. 3, when classifying images, conventional classifiers usually adopt global average pooling (GAP) to generate class score $Y_{\text{hc}} \in \mathbb{R}^{C\times 1\times 1}$ as follows:

$$Y_{\text{hc}} = \frac{1}{|R|}\sum_{(i,j)\in R} y_{c,i,j} \tag{7}$$

where $y_{c,i,j} \in \mathbb{R}^{C\times H\times W}$ is a feature prediction score map with channel number C, height H, and width W. R is the area range of $y_{c,i,j}$, and $|R|$ is the number of pixels contained within $y_{c,i,j}$ area. In the process of generating the class score $Y_{\text{hc}} \in \mathbb{R}^{C\times 1\times 1}$, all pixels are essentially assigned the same class label. To guide the pixel classification, we propose the PAE module, which can adaptively assign a soft label $Y_{\text{sc}} \in \mathbb{R}^{C\times 1\times 1}$ to each pixel to achieve pixel encoding and pixel classification guidance. The PAE module begins by predicting feature score maps using 1×1 convolution. In contrast with the traditional methods, we remove the GAP layer and use the SoftMax function to obtain pixel confidence map $\gamma_{c,i,j} \in \mathbb{R}^{C\times H\times W}$. To produce the final class score Y_{sc}, we regard the pixel confidence map $\gamma_{c,i,j}$ as the weight and calculate the channelwise weighted average of the feature prediction score map $y_{c,i,j}$, as expressed in the following formula:

$$Y_{\text{sc}} = \frac{\sum_{(i,j)\in R}\gamma_{c,i,j}\, y_{c,i,j}}{\sum_{(i',j')\in R}\gamma_{c,i',j'}}. \tag{8}$$

The PAE module successfully transfers the classification supervision from the image level to the pixel level, thereby providing a precondition for explicit constraints on pixel classification. Our method provides sufficiently strong constraints for the classifier to overcome the drawbacks of previous classifiers that only provide image-level supervision and greatly avoid the phenomenon of ultrarange activation in the background region of CAMs.

D. Pixel Global Awareness

In RS images, the foreground and the background are extremely unbalanced (the proportion of background pixels is high). Existing techniques extend the initial activation region

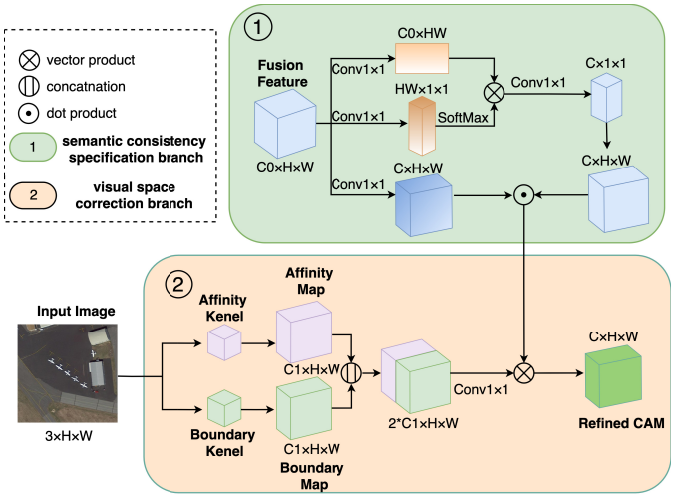


Fig. 4. Architecture of the PGA module consists of two branches: the SCS branch (the top row) and the VSC branch (the bottom row). H, W, and C/C0/C1 denote the height, the width, and the channel numbers of feature maps, respectively.

in CAMs only by expanding the reception field [42] or by adversarial erasing [38] and so on. The above methods ignore the structural and semantic consistency of the target and, therefore, tend to introduce redundant background information and incorrect interclass information. The consequence is that there is a significant mismatch between the edges of the activated regions in CAMs (hereafter referred to as "semantic edges") and the target edges on the image (hereafter referred to as "geometric edges"). To solve the above problem, we propose the PGA module to align the semantic boundary with the geometric boundary and maintain the ideal target response shape of CAMs in complex RS scenes.

As shown in Fig. 4, the PGA module consists of two branches: the semantic consistency specification (SCS) branch (the top row) and the visual space correction (VSC) branch (the bottom row). These two branches facilitate the balance of the activation intensity of the overactive and underactive areas in CAMs and improve the fine-grained information of CAMs. These two branches are described in detail in the following.

1) SCS Branch: The SCS branch uses self-attention to obtain semantic consistency of high-level features, which can be defined as

$$\hat{m}_{:,i} = \frac{1}{\mathcal{C}(\mathbf{x}_i)}\sum_{j=1}^{N_p} f\left(\mathbf{x}_i, \mathbf{x}_j\right) \odot \Psi(m_{:,j}) \tag{9}$$

where i is the spatial position index and j is all possible locations near i. The linear transformation operation Ψ is implemented by 1×1 convolution layers, and the cosine function $f(\mathbf{x}_i, \mathbf{x}_j)$ is utilized to evaluate the similarity between i and j. Finally, the output signal is normalized by $\mathcal{C}(\mathbf{x}) = \sum_{j=1}^{N_p} f(\mathbf{x}_i, \mathbf{x}_j)$.

Nevertheless, in our early experiments, the traditional self-attention structure could easily lead to memory explosions during the refinement of CAMs. As shown in Fig. 5(a), the traditional self-attention structure generates feature maps with the feature dimensions of HW $\times$ HW. Once the input

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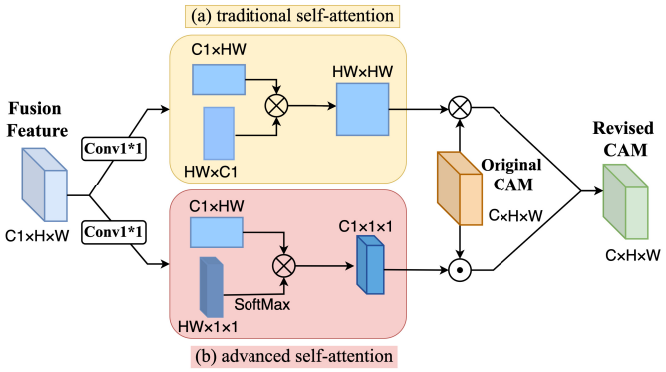


Fig. 5. Comparison of different self-attention structures: (a) traditional self-attention and (b) our advanced self-attention. The feature dimensions are shown as $C \times H \times W$, which denotes a feature map with channel number C, height H, and width W. $\otimes$ denotes the matrix multiplication, and $\odot$ denotes the elementwise multiplication.

image size is slightly increased, the feature dimensions greatly expand. To alleviate this problem, as shown in Fig. 5(b), we propose an advanced self-attentive structure based on [46] with some modifications. Instead of matrix multiplication, we use elementwise multiplication, which is more in line with the idea of assigning appropriate weights to each element. The feature dimension of the output feature map is $C \times 1 \times 1$, which relates only to the size of channel number C and not to the image size.

2) VSC Branch: After completing the SCS for CAMs, the CAMs $\hat{m}$ refined in the previous step are fed into the VSC branch to further modify the shape and spatial pixel affinity of the CAMs. In particular, a boundary kernel and an affinity kernel are proposed, and an edge attention map $\beta_{:,i,j}$ and an affinity attention map $\alpha_{:,i,j}$ are generated using adaptive convolution [47] with these two novel kernels. These two attention maps then serve to iteratively improve the granularity of the CAMs $\hat{m}$. At the tth iteration, the refined CAMs $\hat{m}$ are expressed as follows:

$$\hat{m}^t_{:,i} = \sum_{(j)\ \in \mathcal{N}(i)} \mathrm{Cat}(\beta_{:,i,j}, \alpha_{:,i,j}) \odot \hat{m}^{t-1}_{:,j} \tag{10}$$

where $\mathrm{Cat}(x, y)$ represents the channelwise concatenation operation of two input vectors x and y. The target edge pixel information is extracted by pixel adaptive convolution with the Laplacian kernel as the boundary kernel to obtain the edge attention map $\beta_{:,i,j}$. In addition, to obtain semantic consistency information, the affinity kernel $D(I_{i,j}, I_{x,y})$ as the convolution kernel of pixel adaptive convolution is employed to evaluate the affinity relationship between pixels and generate the affinity attention map $\alpha_{:,i,j}$. In this article, the affinity kernel $D(I_{i,j}, I_{x,y})$ is obtained by calculating the normalized Euclidean distance between the pixels at (i, j) and (x, y) as

$$D(I_{i,j}, I_{x,y}) = \frac{|I_{i,j} - I_{x,y}|}{\sigma^2} \tag{11}$$

where σ is utilized to normalize the Euclidean distance between pixels based on the standard deviation of the pixel intensity in the image. With the edge and semantic consistency information add to CAMs, the issue of semantic edges in CAMs not matching target geometric edges is resolved, and the quality of the semantic masks in RS images is greatly improved.

E. Dynamic Multiscale Fusion

The scale of different classes of targets in RS scenes varies greatly, and it is not enough to rely on high-level features to generate CAMs. A large number of small targets and dense targets cannot be accurately located. Motivated by this, a large number of high- and low-level feature fusion methods [48], [49], [50], [51] have been proposed to enhance the original feature representation. Although these algorithms are proven to be effective in a large number of experiments, the fusion process is complicated with many manual design components. To make feature fusion a self-learning process, we aim to explore a dynamic and adaptive fusion method.

To this end, a dynamic feature fusion technique with a gating mechanism is proposed to combine high-level features X_h and low-level features X_l for classification. First, a learnable gate $G_d \in \mathbb{R}^{H_l \times W_l}$ is used to randomly generate gate maps with the same sizes as the high- and low-level features. Before fusion, a complementary adjustment is made between the high- and low-level features, which can suppress redundant information and reduce noise while preserving the most salient features. In particular, the category and semantic information of high-level features are added to the low-level features to produce $\hat{X}_l$. Similarly, low-level spatial distributions could be used as extra information to obtain $\hat{X}_h$ in order to enhance the spatial information of high-level features. This complimentary adjustment strategy is expressed as the following two formulas:

$$\hat{X}_l = \delta(X_h) \odot X_l \tag{12}$$

$$\hat{X}_h = \delta(X_l) \odot X_h \tag{13}$$

where $\hat{X}_l$ and $\hat{X}_h$, respectively, correspond to the adjusted low- and high-level features. δ is the SoftMax function for generating high- and low-level attention maps, and $\odot$ denotes the elementwise multiplication. After the complimentary adjustment of high- and low-level features, $\hat{X}_l$ and $\hat{X}_h$ are fed into the following dynamic feature fusion process:

$$X_{hl} = \mathrm{ReLU}(G_d \odot \hat{X}_l + (1 - G_d) \odot \hat{X}_h) \tag{14}$$

$$G_d(i, j)^{H_l \times W_l} = (1 - \eta) \odot \Phi(\mathbf{1}^{H_l \times W_l}, \eta) \tag{15}$$

where X_{hl} represents the fused feature, G_d is the gate map, and $G_d(i, j) \in [0, 1]$ denotes the gate weight of pixel (i, j). The function Φ with the parameter $\eta \in [0, 1]$ is employed to generate a random gating map. During training, the initial value of G_d is a vector of $\mathbf{1}^{H_l \times W_l}$ (containing only 1) with the same size as X_{hl}. In our experiments, Φ is the dropout function, which can set some elements of the vector $\mathbf{1}^{H_l \times W_l}$ to 0 at random, and η is a trainable parameter with an initial value of zero.

F. Improved Loss Function

This section describes in detail all the loss functions associated with our S^{2}EPC network. The overall loss function $\mathcal{L}$ is

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first introduced as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{ecr}} + \mathcal{L}_{\text{psc}} \tag{16}$$

where the classification loss $\mathcal{L}_{\text{cls}}$ serves to localize objects roughly, and the equivariant cross-regularization (ECR) loss $\mathcal{L}_{\text{ecr}}$ is used to constrain the equivariant consistency prediction ability of the network. The PSC loss $\mathcal{L}_{\text{psc}}$ is a novel pixel-level loss utilized to train a classifier to classify pixels.

1) CLS Loss: Since only image-level classification labels are available during the training phase, the proposed method employs the same multilabel soft-margin (MLSM) classification loss ℓ_{cls} as previous work [8], [9], [10] to train the network. The MLSM classification loss is defined as follows:

$$\ell_{\text{cls}}(y, l) = -\frac{1}{C}\sum_{c=1}^{C} l_c \log\left(\frac{1}{1+e^{-y_c}}\right) + (1-l_c)\log\left(\frac{e^{-y_c}}{1+e^{-y_c}}\right) \tag{17}$$

where $l \in \{0,1\}^C$ denotes the image category label. In the previous methods, the GAP layer was used to obtain the prediction vector $y_c \in \mathbb{R}^{C\times 1\times 1}$. However, the GAP only provides guidance for target positioning and not pixel classification, which is not conducive to generating high fine-grained CAMs. Instead of employing the traditional equal-weighted average, the PAE module can compute the pixel-weighted average of the feature prediction score map $y_{c,i,j}$ with the pixel confidence map $\gamma_{c,i,j} \in \mathbb{R}^{c\times h\times w}$ as the weight.

With our two-branch S^{2}EPC network, the original image I_0 and the scaled image I_1 can generate two class scores, y_0 and y_1. The overall classification loss $\mathcal{L}_{\text{cls}}$ of our two-branch network is calculated as follows:

$$\mathcal{L}_{\text{cls}} = \frac{1}{2}(\ell_{\text{cls}}(\mathrm{y}_0, l) + \ell_{\text{cls}}(\mathrm{y}_1, l)). \tag{18}$$

2) Pixel Soft Classification Loss: To further improve the fine granularity of CAMs, the original CAMs, the fused features, and the input images are simultaneously fed into PGA for refinement. During the training process, PAE enables pixel encoding to generate class scores by pixel-weighted average, but this is only the first step in providing explicit constraints on pixel classification. Therefore, we introduce PSC loss $\mathcal{L}_{\text{psc}}$, a novel pixel-level loss based on the cross-entropy loss function that can calculate the pixel-level loss of the original CAMs and the PGA-optimized CAMs. We describe PSC loss $\mathcal{L}_{\text{psc}}$ as follows:

$$\mathcal{L}_{\text{psc}} = -\frac{1}{K}\sum_{n=1}^{K}\sum_{i,j}^{W,H}\left(\mu_{c,i,j}\log\left(m^n_{c,i,j}\right) + \left(1-\mu_{c,i,j}\right)\log\left(1-m^n_{c,i,j}\right)\right) \tag{19}$$

where K is the number of pixels, c denotes the categories, and (i, j) represents the position of each pixel. The PGA-optimized CAMs and the original CAMs are denoted as $\mu_{c,i,j} \in [0,1]$ and $m_{c,i,j} \in [0,1]$, respectively. To prevent CAMs' degeneration, we cut off gradient back-propagation of PGA during refinement. $\mathcal{L}_{\text{psc}}$ can provide effective guidance for pixel classification during training so that the CAMs can better fit the target segmentation mask.

3) Equivariant Cross-Regularization Loss: Moreover, to further improve the consistency of our S^{2}EPC network output, we refer to the ECR loss in the SEAM [8] network, which can be represented as

$$\mathcal{L}_{\text{ecr}} = |T(m_{c,i,j}) - \mu_{c,i,j}| + |T(\mu_{c,i,j}) - m_{c,i,j}|. \tag{20}$$

$T(.)$ is the scale transformation operation that can appropriately scale the obtained CAMs $m_{c,i,j}$ and PGA-optimized CAMs $\mu_{c,i,j}$. With the help of scale transformation, we use the absolute error to measure the loss of two branches and try to minimize it during training.

IV. Experiments

To assess the effectiveness of essential parts of our algorithm, we conduct numerous experiments on the common datasets: ISPRS Vaihingen [16] and iSAID [15]. In addition, we also perform experiments on the PASCAL VOC 2012 [17] and compare it with more advanced methods. Experiment results show that our method is also competitive in NS.

A. Datasets and Evaluation Metrics

1) iSAID: iSAID [15] is a public, large-scale dataset used for the segmentation of aerial imagery. Most images are collected from Google Earth, which are mainly taken by the JL-1 and GF-2 satellites. iSAID contains 15 target categories and one background (BG) category. The target categories specifically include plane (PL), baseball diamond (BD), ship (SH), ground-track field (GTF), basketball court (BC), soccer ball field (SBF), roundabout (RA), storage tank (ST), bridge (BR), large vehicle (LV), tennis court (TC), helicopter (HC), harbor (HA), small vehicle (SV), and swimming pool (SP). In our experimental results, we use abbreviations in brackets rather than the full names of the categories. According to the original division of the dataset, there are 1411 high-resolution images for training and validation; the remaining 458 images are used as the test set. Furthermore, we randomly crop six to eight slices of 512 × 512 size from each picture. Finally, we get 7500 training set slices, 1653 verification set slices, and 1315 test set slices. In the training stage, we only use image-level category labels to generate CAMs, while, in the inference stage, we use pixel-level labels to evaluate the quality of pseudolabels.

2) ISPRS Vaihingen: ISPRS [16] is a dataset of airborne images that cover two different scenes: one is a small village called Vaihingen with a large number of single-family houses, while the other is Postdam, a traditional city with tall buildings and narrow streets. ISPRS not only provides high-resolution orthorectified images but also provides a normalized digital surface model (DSM) generated by the corresponding dense image matching technology. The two different scenes in ISPRS were manually classified into six types of land cover: impervious surfaces, low vegetation, trees, buildings, cars, and clutter. Our experiment mainly be conducted on the Vaihingen dataset. According to the official benchmark, we define 16 images for training and 17 images for testing our model. In our experiments, we only use image-level category tags without DSM.

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TABLE II
COMPARISONS OF CATEGORY PERFORMANCE ON iSAID IN MIOU (%), OA (%), AND F1 (%)

model	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU	OA	F1 score
AffinityNet [10]	77.75	14.70	43.72	3.69	5.77	6.32	18.81	20.40	9.76	7.53	54.25	23.78	42.43	3.95	1.84	1.00	20.82	77.99	30.67
IRNet [9]	76.62	18.88	39.82	6.72	16.62	7.48	24.47	16.35	5.69	22.02	54.91	28.96	29.60	9.81	1.31	3.01	22.64	77.02	33.28
SEAM [8]	73.17	28.35	37.75	4.71	15.31	6.86	43.06	31.40	16.67	19.61	44.30	**31.06**	24.41	7.75	9.81	8.31	24.47	74.30	36.79
SC-CAM [12]	76.23	28.55	36.07	11.11	19.95	5.67	23.07	**36.98**	16.01	24.78	54.32	30.07	29.89	10.29	17.34	11.92	26.62	77.26	42.41
S^{2}EPC	**86.11**	**38.30**	**46.70**	**11.48**	**17.10**	**8.71**	**53.30**	36.68	**34.59**	**31.70**	**56.74**	30.20	**49.10**	**11.59**	**19.00**	**15.86**	**36.79**	**86.05**	**46.91**

3) PASCAL VOC 2012: PASCAL VOC 2012 [17] is frequently used as a baseline dataset for object detection or image segmentation, which involves 20 foreground categories and one background category. Following standard practices [10], [42], additional annotations are obtained from the SBD [52], and we end up with 10 582 images all for training, 1449 images for validation, and 1456 images for testing.

4) Evaluation Metrics: Based on the above datasets, we use the mean Intersection over Union (mIoU) scores, F1 scores, and OA indicators to obtain a quantitative and comprehensive evaluation of our model. According to the convention, we also use true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FN) to express the above three indicators. TP represents the number of positive samples for which the prediction results are positive; FP is the number of negative samples wrongly classified as positive; TN is the number of negative samples accurately classified; and FN denotes the number of positive samples erroneously classified as negative.

Specifically, the mIoU, F1, and OA are defined as follows:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{21}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{22}$$

$$F_1 = \frac{2 \odot \text{precision} \odot \text{recall}}{\text{precision} + \text{recall}} \tag{23}$$

where precision = TP/(TP+FN), recall = TP/(TP+FN), and N is the number of categories. All the indicators mentioned above are the most basic and general standards of measuring the semantic segmentation process. mIoU is the mIoU between ground truth and predicted segmentic masks, which measures the precision rate, and OA measures the recall rate. The F1 score is a composite index of precision rate and recall rate with a maximum of 1 and a minimum of 0. Through a comprehensive comparison of different datasets, we could show relatively more reasonable performance results.

B. Implementation Details

In this article, we adopt SEAM [8] as our baseline model, with ResNet38 [10] as the backbone network. We compare our method with the advanced methods on the above datasets and select a more challenging iSAID dataset for ablation experiments. In the training process, we run the Adam optimizer [53] on a single NVIDIA RTX 3090 GPU with 12 iterations, and the batch size is 32. The learning rate is determined by the polynomial attenuation strategy: $\text{lr}_{\text{itr}} = \text{lr}_{\text{init}}(1-(\text{itr}/\max_\text{itr}))^{\lambda}$, where the initial learning rate is $\text{lr}_{\text{init}} = 0.01$ and $\lambda = 0.9$. During inference, we adopt the multiscale test method to generate pseudosegmentation labels and then fuse these pseudolabels by average. Following previous work [35], [54], [55], we use the denseCRF [44] as a postprocessing step and the DeepLab [56] network to train the segmentation model with pseudolabels as supervised data to further refine the results.

TABLE III
COMPARISONS OF OUR METHOD WITH OTHER SUPERVISION TYPES METHODS USING THE iSAID DATASET. BOUNDING BOX (BB) DENOTES THE BOUNDING BOX-BASED WEAKLY SUPERVISED METHOD; IMAGE TAG (IT) DENOTES THE WEAKLY SUPERVISED METHOD USING IMAGE TAGS

Supervision	model	Backbone	mIoU (%)
full	Deeplab v3 [57]	ResNet50	59.05
semi	SDI [58]	VGG16	54.87
	Song [59]	VGG16	55.15
	JMLNet [59]	ResNet50	56.76
weak (BB)	SDI [58]	VGG16	53.82
	Song [59]	VGG16	54.18
	JMLNet [59]	ResNet50	55.34
weak (IT)	baseline [8]	ResNet38	34.20
	S^{2}EPC	ResNet38	**36.79**

C. Comparative Experiments on SOTA Methods

In this experiment, we compare our method with the SOTA method utilizing the iSAID, ISPRS Vaihingen, and PASCAL VOC 2012 datasets. Our experimental results would are presented in three parts according to different datasets as follows.

1) iSAID Dataset: In this part, we measure the performance of the S^{2}EPC model on the iSAID dataset from three perspectives: the performance comparisons of SOTA methods in the NS domain, the performance comparisons of different supervised methods, and the qualitative results (visualization results) analysis. Details are provided in the following.

a) Compared with SOTA methods: We apply the NS approach to the iSAID dataset for comparison with our own method. The detailed comparison results for each category are shown in Table II. Compared to the baseline, almost all of the CAMs obtained by our S^{2}EPC have a better effect, especially for small and dense targets (such as SV and SH), which have been, respectively, increased by 6.77% and 1.79%. In addition,

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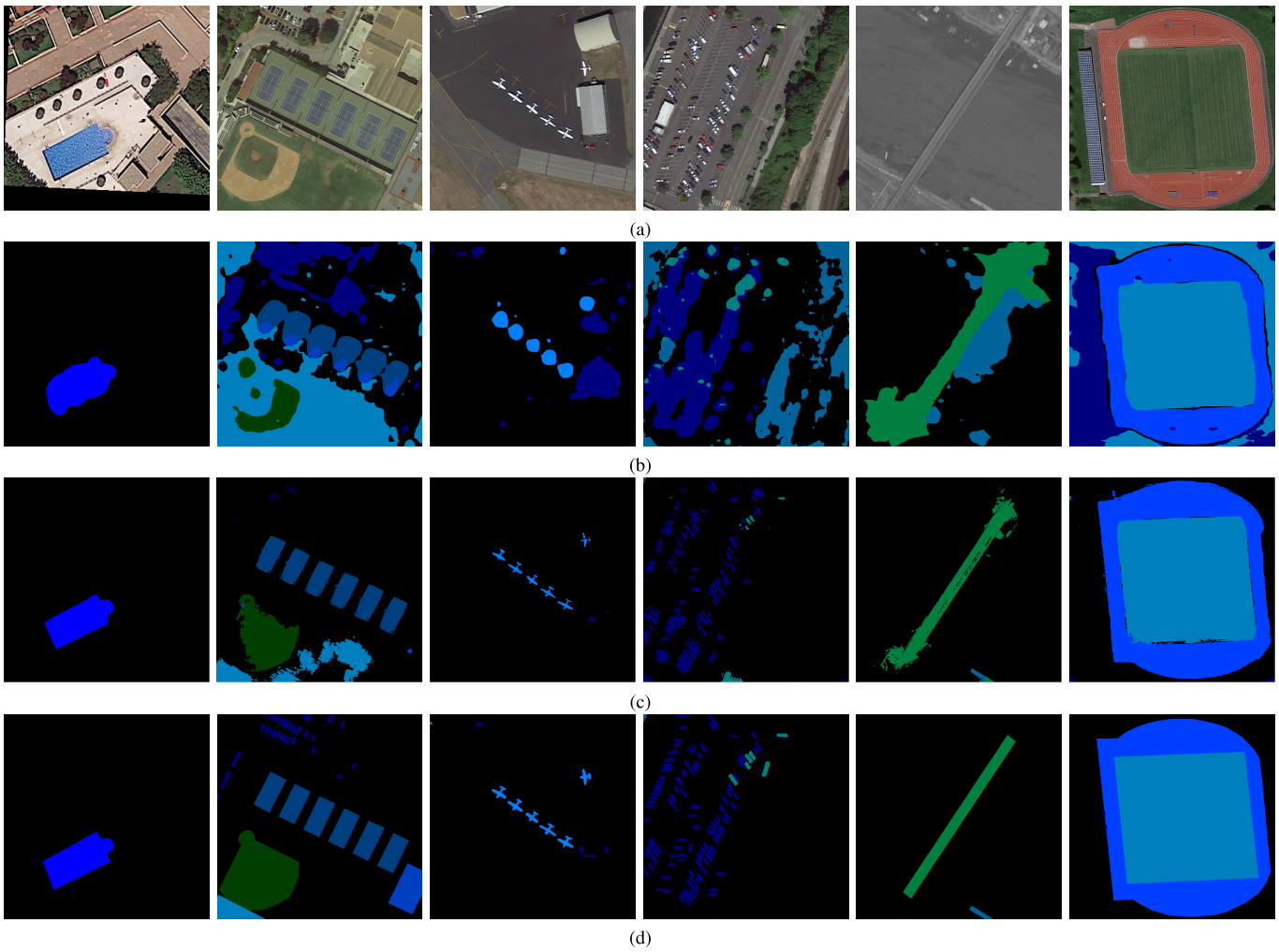


Fig. 6. Visualization of the iSAID dataset results. We show the visualization results of some different sized RS targets, including SVs, PLs, BRs, BDs, BCs, ground track fields, TCs, SBFs, RAs, SPs, and so on. (a) Input image. (b) SEAM. (c) S^2EPC. (d) Ground truth.

TABLE IV
COMPARISONS OF CATEGORY PERFORMANCE ON ISPRS VAIHINGEN IN MIOU (%), OA (%), AND F1 (%)

models	Impervious surfaces	Building	Low Vegetation	Trees	Cars	mIoU	OA	F1 score
AffinityNet [10]	21.45	36.30	28.42	24.91	4.55	23.13	41.06	33.03
IRNet [9]	33.44	43.79	32.55	45.19	6.83	32.36	53.40	41.65
SEAM [8]	35.89	43.71	22.79	51.06	12.12	34.97	53.85	45.87
SC-CAM [12]	44.03	49.80	38.26	49.13	16.92	41.38	54.34	53.49
S^2EPC	**53.86**	**53.43**	**44.40**	**55.79**	**19.66**	**45.43**	**65.57**	**56.33**

many large targets or targets with large size changes, such as BCs and TCs, have been improved to varying degrees of 10.24% and 12.44%, respectively. This is an encouraging result in the field of WSSS.

b) Compared with different supervised methods: Furthermore, we also compare three types of supervision methods: full supervision (Deeplab v3 [57]), semisupervision, and weak supervision (SDI [58], Song [59], and JMLNet [60]) on the iSAID dataset. Our method achieves 36.79% mIoU only by using image tags, as shown in Table III. It is equivalent to 62.3% of the performance of a fully supervised model, 64.8% of the most advanced semisupervised method, and 66.5% of the SOTA weakly supervised method with a bounding box. It appears that this promising result makes it possible for image-level weakly supervised segmentation in the RS field.

c) Visualization results: Fig. 6 shows the qualitative comparison of segmentation results on iSAID between our method and the baseline. With our method, we can obtain more fine-grained segmentation results compared to the baseline. For example, densely distributed SVs, PLs, and STs can be segmented clearly and accurately. The segmentation results of the ground track fields, SBFs, and TCs have more

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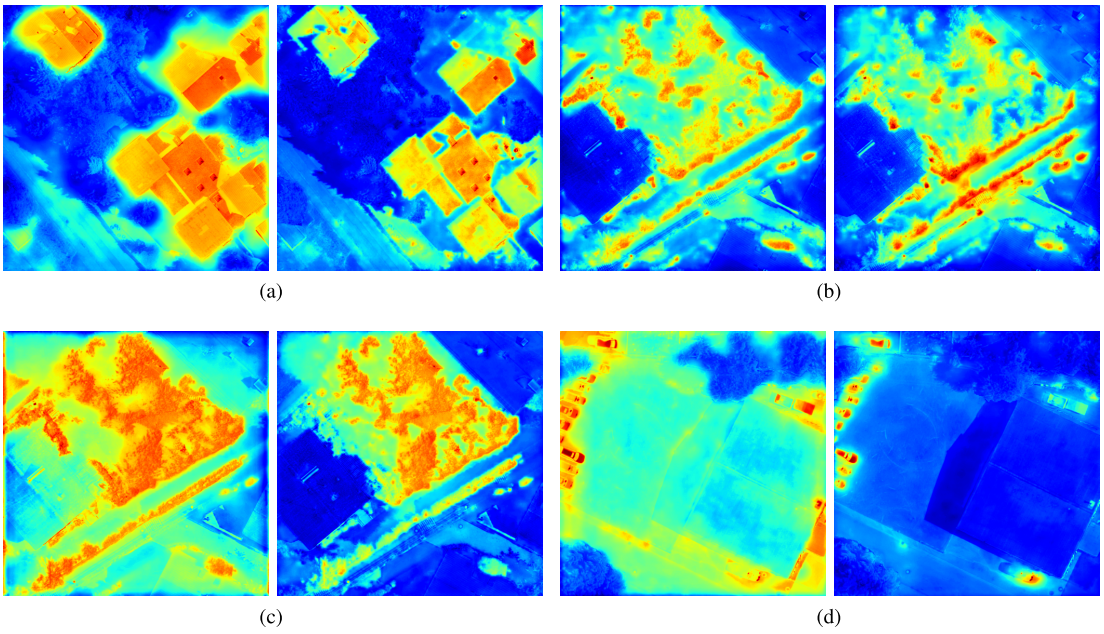


Fig. 7. Visualization of CAMs on ISPRS Vaihingen. (a)–(d) Four categories: building, low vegetation, trees, and cars. (Right) Our results. (Left) Baseline results. (a) Building. (b) Low vegetation. (c) Trees. (d) Cars.

explicit boundaries and more regular shapes. At the same time, the number of misclassified pixels has significantly reduced. It shows that our method has a good balance between semantic consistency and spatial information constraints.

2) ISPRS Vaihingen Dataset: In addition to conducting experiments on the iSAID dataset, we also extend our research to another public RS dataset, ISPRS Vaihingen, to evaluate our method further. Specifically, we evaluate the following two aspects: the performance comparison of SOTA methods in NS and the visualization result analysis.

a) Compared with SOTA methods: In the experiment, we do not consider the "clutter" since it accounts for less than 1% of the total category pixels. We use the same SOTA method as on the iSAID dataset to compare the performance of our model. In Table IV, S^2EPC outperforms the baseline method by far in all six categories of the ISPRS Vaihingen dataset, and the mIoU improves by 10.46% compared to baseline. It is worth mentioning that, for hard samples, such as "Cars," our method can also improve significantly over existing methods.

b) Visualization results: As shown in Fig. 7, we select four categories from the ISPRS Vaihingen dataset for CAM visualization, which are buildings, low vegetation, trees, and cars. It is clear that our method not only provides better location accuracy but also greatly reduces the number of pixels in the background that are incorrectly activated areas, as well as makes the edges of the target look sharper.

3) PASCAL VOC 2012 Dataset: To prove the generalization of our method, we carry out experiments on the PASCAL VOC 2012 dataset and compare it with more advanced methods. Following standard practices, we construct 10 582 images for training, 1449 for validation, and 1456 for testing, and use mIoU as the metric to evaluate segmentation results in the experiments.

TABLE V

COMPARISON ANALYSIS RESULTS WITH OTHER SOTA METHODS ON THE PASCAL VOC 2012 DATASET IN MIOU (%). WE ALSO EVALUATE THE OUTCOMES UNDER DIFFERENT BACKBONE NETWORKS AND MARK THEM WITH DIFFERENT NUMBERS OF SYMBOLS*

Methods	Backbone	Saliency	*val*	*test*
CCNN [61]	VGG16		35.3	35.6
EM-Adapt [20]	VGG16		38.2	39.6
SEC [35]	VGG16		50.7	51.1
STC [62]	VGG16	√	49.8	51.2
AdvErasing [38]	VGG16	√	55.0	55.7
MDC [42]	VGG16	√	60.4	60.8
AffinityNet [10]	ResNet38		61.7	63.7
SEAM [8]	ResNet38		64.5	65.7
Ours	ResNet38		**65.4**	**65.9**
CIAN [63]	ResNet50		62.4	63.8
IRNet [9]	ResNet50		63.5	64.8
Ours*	ResNet50		**67.0**	**67.3**
MCOF [64]	ResNet101	√	60.3	61.2
DCSP [65]	ResNet101	√	60.8	61.9
DSRG [43]	ResNet101	√	61.4	63.2
SeeNet [55]	ResNet101	√	63.1	62.8
BDSSW [66]	ResNet-101		63.6	64.5
OAA [37]	ResNet-101		63.9	65.6
CIAN [63]	ResNet101	√	64.1	64.7
FickleNet [67]	ResNet101	√	64.9	65.3
Ours**	ResNet101		**67.6**	**67.8**

Compared With SOTA Methods: As shown in Table V, our method produces state-of-the-art results on the PASCAL VOC 2012 dataset. Specifically, our S^2EPC pushes the mIoU to 65.4% for the validation set and 65.9% for the test dataset

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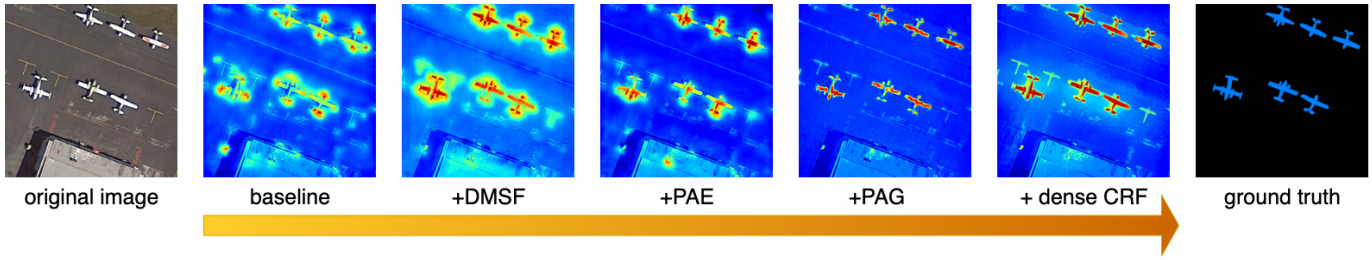


Fig. 8. Visualization results of CAMs on the iSAID dataset after different modules are added to the baseline. The addition of the DMSF module can better locate targets; the PAE module and the PGA module can enhance the fine granularity of CAMs. On the basis of the three modules, the dense CRF operation further enhances the contour and texture of the targets, thus better meeting segmentation mask requirements.

TABLE VI

ABLATION RESEARCH FOR EACH PART OF S^{2}EPC. WE ADD THE PAE, DMSF, AND PIXEL GLOBAL PERCEPTION (PGA) MODULES, IN TURN, TO OBSERVE THE SEGMENTATION PERFORMANCE OF THE MODEL RELATIVE TO THE BASELINE. WE ALSO TEST THE EFFECT OF DENSE CRF AS THE POSTPROCESS STEP ON MODEL PERFORMANCE

baseline	PAE	DMSF	PGA	dense CRF	mIoU (%)
√					22.61
√				√	24.47
√	√				25.69
√	√	√			26.74
√	√	√	√		34.20
√	√	√	√	√	36.79

with resnet38 as the backbone. We also evaluate the outcomes under different backbone networks and mark them with different numbers of symbols * in Table V. We find that the performance of the S^{2}EPC network improves as the number of backbone layers increases, but the corresponding model complexity increases as well. The above results show that our method can also improve the semantic segmentation performance in NS and demonstrate a certain degree of generalizability.

D. Ablation Experiments

In this section, we systematically discuss the effectiveness of the PAE module, the PGA module, and the DMSF module. All of the experimental results presented here are based on the more challenging iSAID dataset.

a) Baseline setup: To make the ablation experiments more rigorous, we still utilize SEAM [8] as the baseline and mIoU as an evaluation metric to evaluate the component effectiveness of each component in the framework. To further refine predictions, we also test the baseline with dense CRF. The results indicate that dense CRF improves the mIoU of the baseline to 22.61%, which is less than our result of 34.20%. After adding dense CRF as the postprocess, our model can further improve the performance up to 36.79%. The detailed results are shown in Table VI.

Compared to the baseline, the overall performance of our network has been greatly improved. The PAE module has increased by 4.87%, and the validity of pixel encoding supports our new idea of realizing classification from the perspective of segmentation. On this basis, to further improve the positioning of small and dense elements, we add the DMSF module and increase mIoU by 0.95%. A dynamic fusion of high- and low-level features can help balance semantics and spatial information, restore details lost in high-level features, and provide more accurate positioning than using only high-level features. Finally, we introduce the PGA module to enhance the RS scene segmentation effect by mining the target structure information and semantic consistency, achieving the best performance of 34.20% on the iSAID dataset. It can be seen that the performance of our model can even reach 36.79% if we add the dense CRF.

Aiming to show the effects of our modules more intuitively, we add PAE, DMSF, and PGA modules to the baseline in turn. The visualization results of CAMs with different stages are shown in Fig. 8. When we input an airport picture with some PLs, the baseline network without any modules initially has difficulty locating the PLs. Following the addition of the DMSF module, all PLs are more accurately positioned, and the activation area of CAMs has been expanded. With the addition of the PAE module, the T-shaped parking spaces without any aircraft are no longer highlighted. Finally, the PGA module is used to align the semantic boundaries of the PLs with the geometric boundaries to obtain more precise and clear semantic boundaries.

b) Effects of PAE: According to the ablation experiments in Table VI, the PAE module improves the mIoU overall by 4.87% compared to the baseline. In addition, further examination of Table VII reveals that the PAE module is capable of producing more fine-grained CAM results regardless of the target size. Specifically, the three larger scale categories of TCs, STs, and BCs increased by 9.51%, 10.56%, and 8.36%, respectively. SHs and SVs also increased by 7.28% and 1.09%. To provide a more intuitive sense of the pixel encoding effect, we show the visualization results of CAMs before and after PAE in Fig. 9.

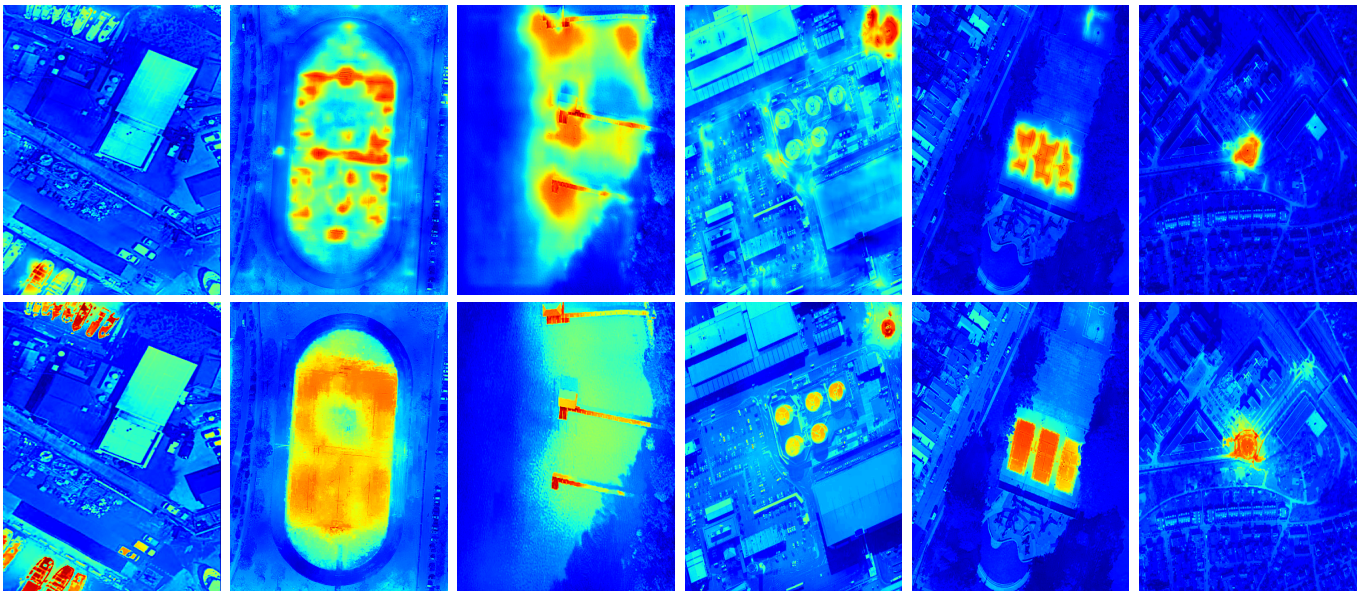
c) Effects of PGA: The PGA module is made up of two parts: the SCS branch and the VSC branch. In the first part, we discuss the effect of higher-level feature selection on the performance of SCS branches. We design channelwise cascade experiments with various combinations of single-layer and multilayer features for the high-level features: Conv4,

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TABLE VII

PERFORMANCE COMPARISONS OF THE BASELINE AND BASELINE + PAE (ABBREVIATED BASELINE*) ON THE iSAID DATASET

model	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU (%)
baseline	73.17	28.35	37.75	4.68	8.03	**6.86**	21.70	31.40	16.67	19.61	44.30	24.89	24.41	7.75	7.80	4.36	22.61
baseline*	**74.57**	**28.55**	**44.21**	**5.77**	**15.31**	6.79	**30.06**	**36.13**	**21.49**	**20.49**	**53.81**	**31.06**	**34.97**	**10.89**	**9.81**	**8.31**	**25.69**



(a) (b) (c) (d) (e) (f)

Fig. 9. Visualization of the PAE module. We selected six categories of different sizes, shapes, and distributions for presentation: SH, SBF, HA, ST, TC, and RA. The upper and lower rows indicate the effect without PAE and with PAE, respectively. (a) SH. (b) SBF. (c) HA. (d) ST. (e) TC. (f) RA.

TABLE VIII

CASCADING RESULTS OF THE SCS BRANCH IN A COMPARISON OF SINGLE-LAYER AND MULTILAYER FEATURES (CONV4, CONV5, AND CONV6)

High-level features	mIoU (%)	Inference times (s)
conv6	20.82	1.171
conv4	23.11	1.243
conv5	23.76	1.469
[conv4,conv5]	30.71	1.592
[conv4,conv5,conv6]	27.70	1.644

TABLE IX

EFFECT OF DIFFERENT NUMBERS OF ITERATIONS ON THE PERFORMANCE OF VSC BRANCHES. AS THE NUMBER OF ITERATIONS INCREASES, THE MODEL INFERENCE TIME GETS LONGER

Number of iterations	mIoU (%)	Inference times (s)
3	26.64	1.407
5	27.34	1.527
10	30.71	1.595
15	29.07	1.654
20	29.97	1.706

Conv5, and Conv6. As shown in Table VIII, multilayer feature cascading yields better results, which improves the semantic consistency of the features. Among them, the combination of Conv4 and Conv5 works best. Therefore, we only use Conv4 and Conv5 channelwise cascading features to complete the remaining parametric ablation experiments.

In the VSC branch, the fine granularity of CAMs is iteratively improved by means of pixel adaptive convolution. In the second part, we mainly discuss the impact of iterations on the performance of the VSC branch. Specifically, we set up a set of experiments with iterations of 3, 5, 10, 15, and 20 to explore the balance between efficiency and performance. The results in Table IX tell us that an excessive number of iterations is not always desirable. The model inference time will get longer as the number of iterations increases, but the ability to constraint the spatial information of CAMs has reached its maximum level. In the end, we keep the setting of iteration number 10, in which our model can get an optimal result with a compromise between inference speed and spatial constraint. A more intuitive visualization effect is placed in Fig. 10. The PGA-optimized CAMs obtain a more accurate pixel classification effect and achieve pixel-level refinement.

d) Effects of DMSF: The DMSF module is designed to adaptively balance high-level semantics and low-level spatial information, especially to improve the localization accuracy of small and dense targets within CAMs. First, we verify

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TABLE X

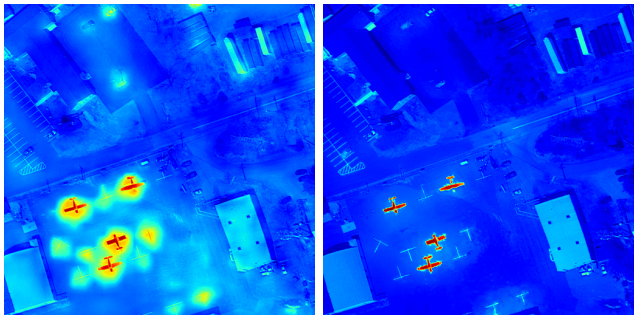
EFFECT OF HIGH- AND LOW-LEVEL FEATURE COMBINATIONS ON DYNAMIC FUSION PERFORMANCE. WE HAVE SOME SYMBOLS THAT NEED TO BE EXPLAINED HERE, THE VERTICAL LINE "|" IS USED TO SEPARATE HIGH- AND LOW-LEVEL FEATURES, THE LEFT-HAND SIDE OF THE VERTICAL LINE IS THE LOW-LEVEL FEATURE, AND THE RIGHT-HAND SIDE IS THE HIGH-LEVEL FEATURE

Select features for fusion	mIoU (%)	inference time (s)
[Conv3|Conv6]	**30.62**	**1.368**
[Conv3|Conv4,Conv5,Conv6]	27.01	1.386
[Conv2,Conv3|Conv6]	30.07	1.398
[Conv2,Conv3|Conv4,Conv5,Conv6]	28.16	1.406

TABLE XI

EVALUATION OF THE DYNAMIC GATING MECHANISM. BASED ON THE ABOVE FEATURE FUSION EXPERIMENT, WE PICK THE BEST COMBINATION OF CONV3 AND CONV6 FOR FEATURE FUSION EXPERIMENTS AND EXPLORE THE EFFECTIVENESS OF THE DYNAMIC FUSION MECHANISM

fusion mechanism	mIoU (%)	inference time (s)
without gating mechanism	28.33	1.382
Static gating mechanism (set gate=0.3)	29.57	1.381
Dynamic gating mechanism	**30.62**	**1.368**



(a) (b)

Fig. 10. Effects comparison before and after using PGA module. The PGA module can match the semantic area boundaries of the aircraft with the geometric boundaries and significantly improve the semantic mask results of RS images. (a) Before PGA module processing. (b) After PGA module processing.

the effect of selecting different high- and low-layer features for fusion. Our backbone network is ResNet38 [68] with six blocks, in which Conv4, Conv5, and Conv6 are all high-level features with the same resolution, and Conv2 and Conv3 are low-level features with different resolutions (Conv2 has a higher resolution than Conv3). These high-level features are determined via atrous convolution at different dilated rates and, therefore, are able to maintain the same resolution. We set up experiments with different combinations of low-level features represented by "Conv3" and "Conv2 and Conv3" and high-level features represented by "Conv6" and "Conv4, Conv5, and Conv6." According to the results of the experiments in Table X, we find that too much superimposition of high-level features does not assist in the target localization of subsequent CAMs. In fact, some redundant information would be introduced, resulting in negative consequences. Low-level features have more spatial information, which makes it easier to recover the information lost in the high-level features. Therefore, we, finally, choose the feature combination of Conv3 and Conv6 to obtain the best experimental results.

TABLE XII

EVALUATION OF THE COMPLEMENTARY ADJUSTMENT MECHANISM. WE STILL PICK THE BEST COMBINATION OF CONV3 AND CONV6 FOR FEATURE FUSION EXPERIMENTS. X_l AND X_h REPRESENT THE LOW- AND HIGH-LEVEL FEATURES, RESPECTIVELY; $\hat{X}_l$ AND $\hat{X}_h$ BOTH DENOTE THE FEATURES AFTER THE COMPLEMENTARY ADJUSTMENT OPERATION

fusion mechanism	mIoU (%)	inference time (s)
X_l, X_h	29.32	1.326
$\hat{X}_l, X_h$	29.68	1.339
$X_l, \hat{X}_h$	30.21	1.355
$\hat{X}_l, \hat{X}_h$	**30.62**	**1.368**

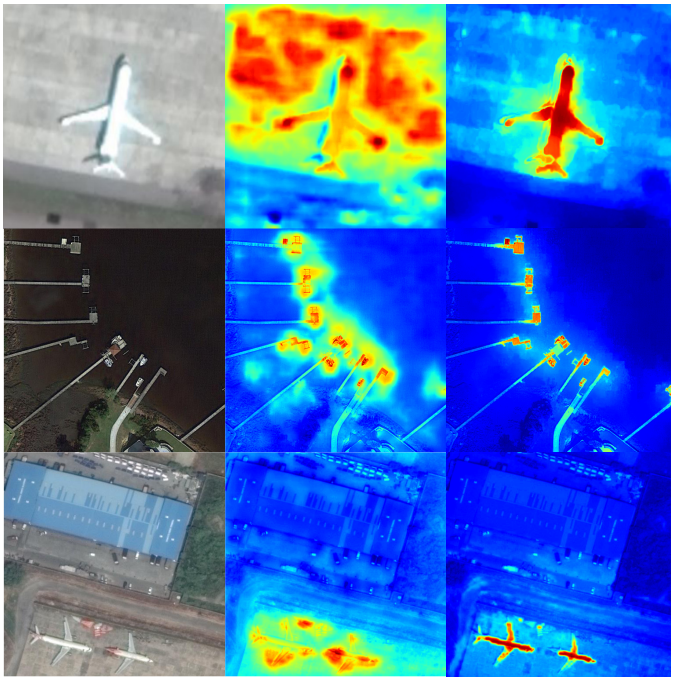
It is noteworthy that we also design a complementary adjustment mechanism for high and low layers that serves to suppress redundant information and retain salient information prior to feature dynamic fusion. We choose the best feature combination of Conv3 and Conv6 to evaluate the effectiveness of the complementary adjustment mechanism. The experimental results are recorded in Table XI. It is obvious that the dynamic gating fusion mechanism has a better performance in both fusion effects and inference speed. As shown in Table XII, the use of the complementary adjustment operation helps to improve the results, especially when the high- and low-level features are complementarily adjusted together to obtain the best results.

In the above experiment, Conv3 and Conv6 are chosen as the best feature fusion method to further validate the effectiveness of dynamic gating fusion. We realize the integration of Conv3 and Conv6 by the following three methods:

1) no gating mechanism;
2) static gating mechanism (the gate value is set to 0.3);
3) dynamic fusion mechanism with gating (the gate value needs to be learned).

As shown in Fig. 11, we visualize the results of the DMSF module before and after processing. Due to the addition of the DMSF module, CAMs only produce highlights in target areas,

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(a) (b) (c)

Fig. 11. Visualization of the CAMs before and after using the DMSF module. DMSF can effectively suppress incorrect activation regions in the background and improve the localization accuracy of targets, especially small targets. (a) Input image. (b) Without DMSF. (c) After DMSF.

which significantly suppresses the incorrect activation areas in the background. The results demonstrate that our method can more reasonably and effectively combine low-level spatial information with high-level semantic information to suppress noise and redundant data.

V. Conclusion

This article presents a novel self-supervised Siamese network framework based on explicit pixel-level constraints (S^2EPC), which is the first benchmark for image-level WSSS tasks in multiclass RS scenes. In particular, the PAE module and the PGA module are designed to enhance the fine granularity of the CAMs so that CAMs can better fit target segmentation masks. Furthermore, the DMSF module with a gating mechanism is utilized to improve the positioning accuracy of small and dense targets. Extensive experiments confirm the rationality and effectiveness of our method in RS scenes and even in NS scenes. In future work, we intend to continue improving model performance to close the gap between weakly supervised segmentation and fully supervised segmentation.

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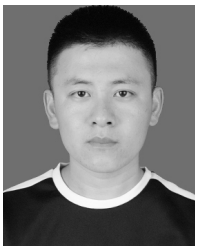
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